The holistic storage of verb+*up* phrases in text-based and audio-based language models

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Abstract

Humans both compute novel forms using abstract representations and store frequently encountered items as holistic units, but the factors governing this tradeoff remain poorly understood. While recent work has examined abstract knowledge in language models, holistic storage of multi-word units has received far less attention. In the present study, we probe internal representations in text-based LLMs and an ASR model, testing whether V+*up* phrasal verbs develop distinct representations as a function of frequency and predictability. Both model types exhibit holistic storage effects, with predictability effects strengthening with scale. These results support usage-based accounts and suggest that error-driven learning produces human-like storage patterns across modalities.

1 Introduction

A central debate in linguistics concerns how humans trade off between computation and storage (Houghton, 2025a,b; Houghton and Morgan, 2023, 2024; Kapatsinski, 2010; Morgan and Levy, 2015, 2016, 2024; Stemberger and MacWhinney, 2004, 1986). Computation refers to applying abstract knowledge to generate new representations; for example, deriving *wugs* from *wug* via a productive pluralization rule. Storage refers to retrieving a whole representation from memory rather than computing it, as when a common phrase is accessed as a unit even though it could in principle be derived compositionally. Both mechanisms are clearly at work in language learning and processing, but the factors that drive what forms are produced via computation and what forms are stored and retrieved holistically remains poorly understood.

1.1 Computation vs Storage in Humans

There is an abundance of evidence for abstract knowledge in language use by humans. Children generalize morphological rules productively to novel words they have never encountered (Berko, 1958), and priming studies show that sentences sharing syntactic structure or semantically related words facilitate each other's processing, implying that something abstract is shared across their representations (Bock, 1986; Meyer and Schvaneveldt, 1971). Similarly, when ordering novel or low-frequency binomials, humans rely on general abstract preferences rather than item-specific knowledge (Morgan and Levy, 2016).

Storage has had a more controversial history. Early accounts held that only irregular forms were stored holistically and that any compositionally derivable form was derived via computation (Pinker and Ullman, 2002). Evidence against this strong view has since accumulated (Bybee and Scheibman, 1999; Houghton, 2025b; Kapatsinski, 2010; Morgan and Levy, 2016; Stemberger and MacWhinney, 2004). Stemberger and MacWhinney (2004) showed that inflection errors are less common for high-frequency words, suggesting frequency-sensitive storage even for regularly derived forms. Bybee and Scheibman (1999) demonstrated that *don't* is phonetically more reduced in high-frequency phrases like *I don't know* than in lower-frequency phrases like *I don't go*; if *don't* always had the same underlying representation, such phrase-level reduction is difficult to explain.

Processing studies provide converging evidence. Morgan and Levy (2016) found that while low-frequency binomials are ordered by abstract preferences, high-frequency binomials are governed by item-specific preferences, suggesting a frequency-dependent shift from computation to storage. Most directly relevant to the present study, Kapatsinski (2010) found that listeners are slower to detect *up* in high-frequency V+*up* phrases than in lower-frequency ones, and Houghton (2025b) and found that this replicates for phrases varying in predictability as well. These findings suggest that high-frequency and high-predictability V+*up* phrases are represented holistically, with *up* no longer as accessible as a distinct unit.

1.2 Computation vs Storage in Language Models

Whether language models exhibit analogous computation-storage tradeoffs has become an active area of inquiry. On the computation side, results are mixed: some studies find that models learn abstract generalizations not present in training (Misra and Mahowald, 2024; Yao et al., 2025), while others find that models fail where abstract knowledge is required, such as morphological generalization to novel words (Haley, 2020) or relying on item-specific rather than abstract preferences where humans use the latter (Houghton et al., 2025; Lasri et al., 2022). On the storage side, there is no doubt that models rely heavily on memorization (McCoy et al., 2023), and item-specific frequency effects have been documented in tasks where humans show abstract preferences (Houghton et al., 2025).

Whether language models develop holistic phrasal representations in the same way as humans is an open question. If they do, this would suggest that holistic storage is a natural consequence of learning from distributional patterns in the language, requiring no special storage mechanism beyond what is needed for general language learning, and lending support to usage-based accounts of how computation and storage interact. This question is especially pressing for audio-based models: the human behavioral evidence reviewed above comes from listening paradigms. An automatic speech recognition (ASR) model is therefore arguably the most ecologically valid computational analog of the human results. And yet, holistic phrasal storage has never been examined in any such model. More broadly, holistic phrasal storage in language models has rarely been examined in any modality, and it has not been studied in models trained on human-comparable amounts of data.

1.3 Present Study

The present study addresses this gap. We probe internal representations in both text-based language models and an audio-based speech recognition model using classifiers trained to detect the embedding of standalone *up*, then applied to V+*up* phrases of varying frequency and predictability. If these models develop holistic phrasal representations, the representation of *up* in high-frequency and high-predictability V+*up* phrases should diverge from that of standalone *up* more than in lower-frequency and lower-predictability phrases. Our specific contributions are:

- We show that holistic phrasal storage emerges in both text-based LLMs and an ASR model, establishing that frequency- and predictability-driven holistic representations arise from exposure-based learning across modalities, including the auditory domain in which the human behavioral evidence was originally obtained.

- We show that frequency effects on phrasal storage are robust across model sizes, but predictability effects strengthen with scale and emerge in earlier layers in larger models, suggesting that sensitivity to co-occurrence statistics beyond raw frequency requires greater representational capacity.

- We show that Whisper's encoder and decoder exhibit opposing effects: the encoder, primarily responsible for acoustic processing, represents high-frequency V+*up* phrases *more* similarly to standalone *up*, consistent with repeated acoustic exposure anchoring the *up* representation near its standalone form, while the decoder, where linguistic representations are primary, shows holistic storage for both frequency and predictability. This dissociation reveals that in audio models, holistic storage is localized to the decoder.

- We train and release three open-access BabyLM models (Charpentier et al., 2025), checkpointed every 20M tokens, for future research on human-scale language learning.

2 Model Training

We trained three autoregressive language models on the BabyLM corpus (Charpentier et al., 2025), a 150M-token dataset designed to be more reflective of the scale and quality of the language that humans receive.[1] All three models follow the OPT decoder-only transformer architecture (Zhang et al., 2022), with 125M, 350M, and 1.3B parameters, respectively.

[1]Though it is worth noting, as has been pointed out before (e.g., Houghton, 2025b), that it can be misleading to compare the tokens that LLMs receive to the "tokens" that humans receive, since humans encounter language in a context-rich environment while LLMs see only the raw text.

Prior to training, we fit a byte-pair encoding (BPE) tokenizer directly on the BabyLM training corpus, yielding a vocabulary of 8,192 subword types. This tokenizer was shared across all three model sizes, ensuring that cross-model comparisons are not confounded by differences in tokenization. A full description of the model training is included in the Appendix.

3 Experiment 1: UP Independently

Experiment 1 tests whether LLMs develop holistic phrasal representations analogous to those proposed by usage-based theories (e.g., Houghton, 2025b; Kapatsinski, 2010), using classifiers trained on standalone *up* representations and applied to V+*up* phrases varying in frequency and predictability. We investigate this across OLMo-3 7B and the three BabyLMs (OPT-125M, OPT-350M, OPT-1.3B).

3.1 Methods

For each model (OLMo-3 7B and BabyLM OPT-125M, OPT-350M, and OPT-1.3B), we extracted the representation of the token *up* from each hidden layer of the models for each sentence. Using these representations, we trained a separate logistic regression classifier for each layer to distinguish standalone *up* (which did not occur in V+*up* phrases in the training of the classifier) from other tokens in the same sentence (these other tokens did not contain the segment *up* in any capacity); the training items are described below. The trained classifier was then tested on a held-out test set comprising V+*up* phrases, and for each sentence the classifier returned a logit score reflecting how confident the classifier was that the representation of *up* in the V+*up* phrase resembled the standalone class. We then examined the logit returned by the classifier as a function of frequency and predictability of the V+*up* phrase.

Frequency for each V+*up* type is operationalized as its raw corpus count, log-transformed:

$$\text{log-frequency} = \log\big(\text{count}(V\text{+up})\big) \qquad (1)$$

Predictability is operationalized as the log-odds ratio of V+*up* occurrences to V-only occurrences in the corpus. Letting $c_{up} = \text{count}(V\text{+up})$ and $c_V = \text{count}(V)$:

$$\text{log-predictability} = \log\left(\frac{c_{up}}{c_V - c_{up}}\right) \qquad (2)$$

In order to calculate frequency and predictability, we used corpus statistics for the dataset each model was trained on. As a result, the frequency and predictability statistics varied between the BabyLMs and the OLMo model. Specifically, for the BabyLM models, corpus statistics were calculated from the BabyLM V3 dataset (Charpentier et al., 2025). For OLMo-3 7B, they were computed from Dolma v1.7, queried via the infini-gram API (Liu et al., 2024). Although OLMo-3 7B was trained on Dolma 3, a larger, more recent Dolma-derived corpus, Dolma v1.7 is the most recent Dolma snapshot indexed by infini-gram and draws from the same underlying sources.[2]

3.1.1 Classifier Training

The classifier was trained to distinguish the language models' representations of the preposition *up* from representations of other (non-*up*) tokens in the same sentence. Positive training examples consisted of 1,000 occurrences of standalone *up* (i.e., *up* not occurring in a V+*up* phrase, such as its use as a preposition or adverb), drawn from sentences in the C4 corpus (Raffel et al., 2020). Negative examples were 1,000 tokens randomly selected from the same sentences, restricted to tokens whose decoded string consists entirely of alphabetic characters (no numbers, punctuation, or special characters), and excluding the preposition *up* itself and any token containing *up* as a substring. The validation set was drawn from the same pool of sentences (a non-overlapping subset), with token positions resolved per model's tokenizer; exact counts fall slightly below 1,000 for some models because the tokenizer does not always produce an isolated *up* token. For example, when *up* appears before punctuation and the tokenizer fuses the two into a single unit (e.g., *up,*), there is no isolable *up* position to extract, so that instance is excluded.

The classifiers targeting OLMo-3 7B and the BabyLM models were trained on the **same underlying sentences**, but the frequency and predictability statistics used in the analyses were derived from the respective corpus for each model (Dolma v1.7 for OLMo-3 7B, queried via the infini-gram API; and the BabyLM training corpus for the BabyLM models). In other words, the training sentences

[2]The relative frequencies and co-occurrence patterns of common English phrasal verbs are unlikely to differ substantially between Dolma versions, as both are large-scale web-text corpora of similar composition. We therefore treat Dolma v1.7 statistics as a reasonable approximation of OLMo-3 7B's training distribution.

were identical across models, but the tokenization and corpus statistics varied.

Classifier training and validation split sizes are shown in the Appendix (Table 4). The test set comprised verb+*up* phrases (e.g., *pick up*, *set up*) with at least 20 occurrences in the corpus; up to 20 sentences were sampled per type. Corpus frequency and predictability (log odds; see Equation 2) were derived from the corpus described above for each model (Dolma v1.7 for OLMo-3 7B; the BabyLM training corpus for BabyLM models). Because the BabyLM corpus is smaller than Dolma, fewer V+*up* types attain a valid (non-zero) predictability estimate under BabyLM statistics; only types with a valid predictability estimate are included in the analyses. Full item-level statistics are reported in Table 3 (Appendix).

For OLMo-3 7B, 3,927 unique V+*up* types were included, with a median corpus frequency of 92 and a median log-predictability of -5.38. The BabyLM models shared 1,039 items; these items had a higher median frequency (920) and higher median log-predictability (-3.40), reflecting the smaller and more concentrated vocabulary of the BabyLM training data.

3.1.2 Analyses

In order to examine the effects of frequency and predictability on the representations of *up* in V+*up* phrases, we implemented a Bayesian mixed-effects regression model using the *brms* package (Bürkner, 2017). For each statistical analysis, the outcome variable was the logit score returned by the classifier on each test item. We included fixed-effects for frequency and predictability, both of which were centered and scaled (denoted c_log_freq and c_log_predic below). We also included random intercepts for phrasal verb type (since there were multiple sentences for each phrasal verb). Weak, uninformative priors were included on each fixed-effect. The model syntax is included below:

$$\begin{aligned}\text{logit} &\sim c\_\text{log\_freq} \times c\_\text{log\_predic} \\ &\quad + (1 \mid \text{verb\_up})\end{aligned} \tag{3}$$

Separate statistical models were fit for each language model at the final hidden layer. For each coefficient we report the posterior mean, standard error, 95% credible interval, and the proportion of posterior samples greater than zero (% > 0).

In order to examine the difference in representations across hidden layers, we additionally fit a generalized additive model (Wood, 2017). Specifically, two models were fit separately, one for frequency and one for predictability, each with a tensor product smooth over the predictor and hidden layer index, as well as a random intercept by phrasal verb type:

$$\begin{aligned}\text{logit} &\sim \text{te}(\text{log\_freq},\ \text{layer}) \\ &\quad + s(\text{verb\_up},\ \text{bs}=\texttt{'re'}) \\ \text{logit} &\sim \text{te}(\text{log\_predic},\ \text{layer}) \\ &\quad + s(\text{verb\_up},\ \text{bs}=\texttt{'re'})\end{aligned} \tag{4}$$

3.2 Results

Full numerical results are reported in Table 5 for the Bayesian model and Table 6 for the GAM (Appendix); they are visualized in Figure 1 and Figure 2. Figure 8 (Appendix) shows the layer-by-layer difference between high- and low-predictor items more directly.

For all four LLMs, there was a negative effect of frequency on the classifier's logit score. Further, for three of the four models (OLMo-3 7B, BabyLM 125M, and BabyLM 1.3B) there was a negative effect of predictability (though the 125M CI crosses zero, over 96% of posterior samples are negative, which we treat as a meaningful effect). In other words, both frequency and predictability result in the representation of *up* in V+*up* phrases being less similar to the standalone representation of *up*.

Additionally, the by-layer analysis demonstrates that this divergence begins to appear early on in the larger models (specifically, OLMo-3 7B and BabyLM 1.3B show effects relatively early on), while the smaller models do not show an effect until later layers. This suggests that for larger models, the representations of high-frequency V+*up* phrases are more separated from the representations of the individual constituent words (in the case of the present experiment, the representation of standalone *up*) than for smaller models.

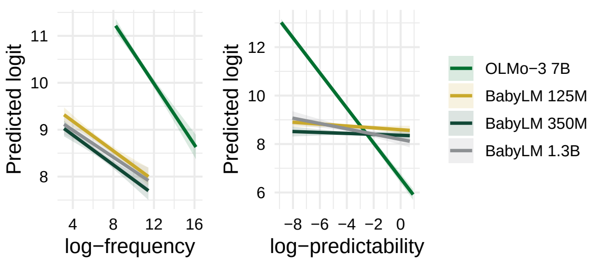

Figure 1: Final-layer brms predicted logit by frequency (left) and predictability (right) for all models (UP independently). Lines slice the joint surface at the other predictor's mean; bands = 95% CIs.

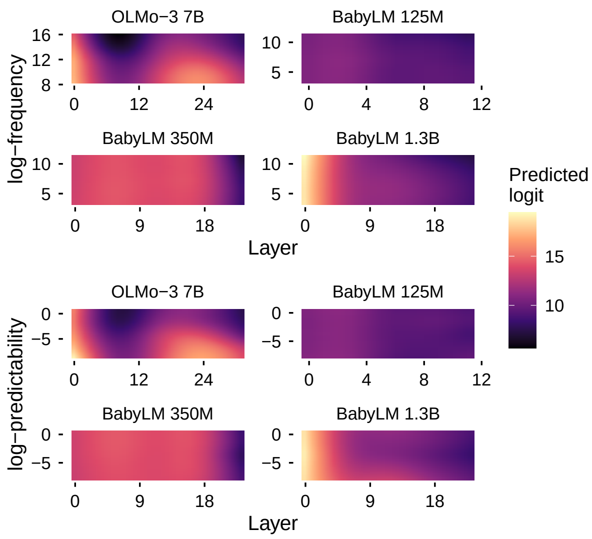

Figure 2: GAM-predicted logit by layer and predictor for all models (UP independently). Top: frequency; bottom: predictability. Color = logit; random effects excluded.

3.3 Discussion

Experiment 1 found that a classifier trained on standalone *up* representations was less confident that high-frequency and high-predictability V+*up* phrases contained the representation of standalone *up* compared to lower-frequency and lower-predictability V+*up* phrases. These results replicate those of Kapatsinski (2010) and Houghton (2025b), who targeted humans' representations. In Experiment 2, we more closely replicate these studies by also examining representations of *up* as a subword unit.

4 Experiment 2: UP as Subword

In Experiment 2, we replicate the design of Experiment 1 with one key modification: the classifier is trained on instances of *up* embedded within a larger word (e.g., *update*, *upon*, etc), requiring it to distinguish the sound segment *up* as opposed to specifically the preposition *up*, from other tokens. This tests whether the frequency and predictability effects observed in Experiment 1 generalize to the sound segment more generally. This is arguably more faithful to Kapatsinski (2010) and Houghton (2025b)'s studies where participants were tasked with recognizing *up* in general, regardless of whether it occurred within a word or as a preposition.

4.1 Methods

The procedure was identical to Experiment 1 (same models, OLMo-3 7B and BabyLM OPT-125M, OPT-350M, OPT-1.3B; classifiers trained and evaluated at each hidden layer; same test set, outcome variable, and statistical approach) with one difference: the classifier was trained on a broader set of positive examples that included instances of *up* embedded within a larger word (e.g., *update*, *upon*).

4.1.1 Classifier Training

The training set for this experiment combined two types of positive example: 1,000 occurrences of standalone *up* (identical to Experiment 1) and 1,000 occurrences of *up* embedded within a larger word (e.g., *update*, *upon*, *setup*). Critically, the subword positives were restricted to **unique word types**: each up-containing word form contributed exactly one instance, so the classifier could not learn to recognise a high-frequency form like *upon* from repeated exposure. Negative examples (1,000 drawn from each sentence pool) were tokens consisting entirely of alphabetic characters that did not contain *up* as a substring. The validation set was constructed by the same procedure (targeting 1,000 positive, 1,000 negative per sentence pool; exact counts vary by tokenizer; see Table 7). As in Experiment 1, OLMo-3 7B and BabyLM models used the same underlying sentences with their own tokenizers. The test set was identical to Experiment 1. Classifier training and validation split sizes are shown in Table 7 (Appendix).

4.1.2 Analyses

Analyses were identical to those of Experiment 1, using the same model formula (Equation 3), predictor definitions (Equation 1, Equation 2), priors, and random-effects structure. The brms models were fit separately for OLMo-3 7B and each BabyLM model at the final transformer layer. Across-layer GAMs were fit using the same approach as Experiment 1 (Equation 4).

4.2 Results

Full numerical results are reported in Table 8 for the Bayesian model and Table 9 for the GAM (Appendix); they are visualized in Figure 3 and Figure 4. Figure 9 (Appendix) shows the layer-by-layer difference more directly. Frequency effects replicate Experiment 1 across all four models. Predictability effects are more variable: the 125M and 350M BabyLMs show weak or absent effects, while the 1.3B and OLMo-3 7B show progressively stronger negative effects, consistent with predictability sensitivity increasing with scale. The by-layer patterns are also more variable than in Experiment 1, with frequency effects generally more stable across layers than predictability effects.

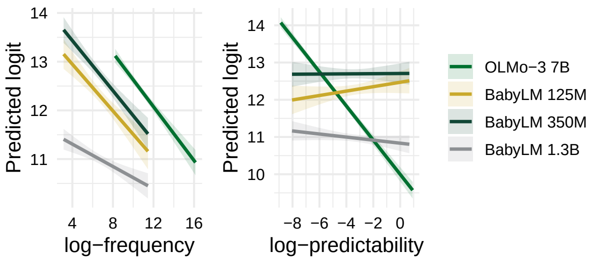

Figure 3: Final-layer brms predicted logit by frequency (left) and predictability (right) for all models (UP as subword). Lines and bands as in Figure 1.

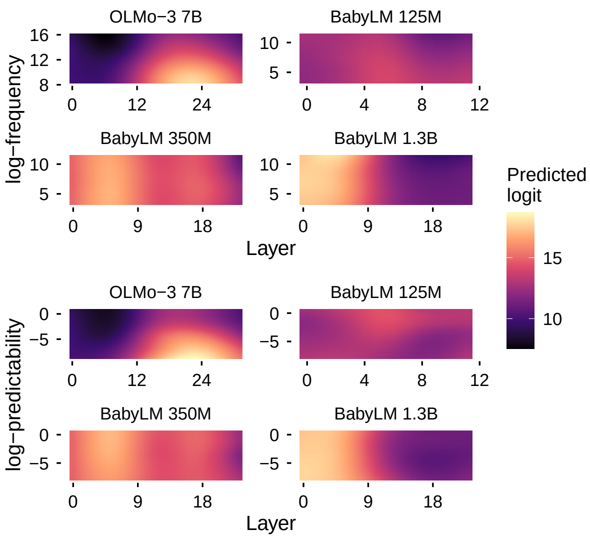

Figure 4: GAM-predicted logit as a function of transformer layer (x-axis) and predictor value (y-axis) for each model (UP as subword). Top row: log-frequency; bottom row: log-predictability. Color encodes the predicted logit score (random effects excluded).

4.3 Discussion

The results of Experiment 2 mostly replicate those of Experiment 1. Frequency shows a strong, consistent effect such that high-frequency V+*up* phrases diverge from representations of *up*, even as a subword unit. The emergence of predictability effects with increasing parameter count suggests that sensitivity to predictability may require more representational capacity than sensitivity to frequency. This is perhaps unsurprising: frequency is directly recoverable from surface co-occurrence counts, while predictability requires tracking the conditional distribution of the particle given the verb — a more relational statistic that may only become reliably encoded at larger scale.

5 Experiment 3: Whisper (ASR Model)

In Experiment 3, we apply the same classifier approach to Whisper-small, an automatic speech recognition (ASR) model trained on spoken audio rather than written text. Whisper differs from the models in Experiments 1 and 2 in both its training modality and its encoder-decoder architecture. Examining Whisper allows us to ask whether frequency and predictability effects on phrasal representations generalize beyond text-based models to representations learned from speech, and whether such effects differ between the encoder (which processes acoustic input) and the decoder (which generates text output).

5.1 Methods

The procedure mirrored Experiments 1 and 2, with two key differences: (1) the model was Whisper-small, an ASR model with an encoder-decoder architecture trained on spoken audio rather than text; and (2) the stimuli were spoken audio segments rather than written sentences. For each segment, we ran the audio through Whisper and extracted the hidden-state representation of *up* at each layer of the encoder and decoder separately. A logistic regression classifier was trained independently for each component (encoder, decoder) on spoken instances of standalone *up* (see Classifier Training below), then applied to spoken V+*up* phrasal verb segments. As in Experiments 1 and 2, each segment contributed one observation, with random intercepts by phrasal verb type. For the sake of brevity, we only trained the classifier to differentiate between Whisper's embeddings of *up* as a standalone preposition and non-*up* tokens.

5.1.1 Classifier Training

The Whisper experiment used spoken language data drawn from a mixed-source audio metadata file (`up-audio-metadata.csv`) containing 224,118 timestamped speech segments in which *up* occurred. Word-level timestamps within each segment were obtained using WhisperX forced alignment. Each occurrence of *up* was classified by running spaCy part-of-speech tagging on the full segment transcript: if the token immediately preceding *up* was tagged as a VERB, the instance was labelled *V+up* (e.g., *pick up*, *clean up*). This classification procedure mirrors the one used to identify standalone *up* in Experiment 1. Of the 224,118 segments, 165,367 were classified as *V+up*, spanning 4,161 unique V+*up* types.

The **training and validation sets** used 1,000 standalone *up* instances each as positive examples, paired with 1,000 randomly selected non-*up* words from the same segment as negative examples. This mirrors the standalone-*up* training design of Experiment 1. Negative words were restricted to those whose string consists entirely of alphabetic characters (matching the `token.isalpha()` criterion used in Experiments 1 and 2), with a minimum duration of 20 ms. Training and validation segments were non-overlapping.

The **test set** comprised V+*up* types with at least 5 occurrences in the audio dataset (lower than Experiments 1 and 2, reflecting sparser audio coverage), with up to 20 instances per type. Corpus frequency and predictability were drawn from Dolma v1.7 (via infini-gram), identical to OLMo-3 7B. After filtering to types with a valid predictability estimate, 1,397 unique V+*up* types were retained (median corpus frequency: 683; median log-predictability: -4.05). Split sizes and item-level statistics are in Table 10 and Table 3 (Appendix).

5.1.2 Analyses

Analyses followed the same procedure as Experiments 1 and 2, using the same model formula (Equation 3) and predictor definitions (Equation 1, Equation 2). The brms models were fit separately for Whisper's encoder and decoder at the final transformer layer. Corpus statistics were drawn from Dolma v1.7 (queried via the infini-gram API) for both components. Across-layer GAMs were fit using the same approach as Experiment 1 (Equation 4).

5.2 Results

Full numerical results are reported in Table 11 for the Bayesian model and Table 12 for the GAM model (Appendix); they are visualized in Figure 5 and Figure 6 below. Figure 10 (Appendix) shows the layer-by-layer difference between high- and low-predictor items more directly.

The results for Whisper show a different trend between the encoder and decoder. For the encoder, the effect of frequency on classifier logits is positive while the effect of predictability is negative. On the other hand, for the decoder, the effects of both frequency and predictability are negative. Further, there is also a negative interaction between frequency and predictability in the decoder, suggesting that the negative effect of frequency is even *more* negative when the V+*up* phrase is also predictable.

The by-layer analysis demonstrates that the effect of frequency in the encoder actually starts out as *negative* and only becomes positive in later layers of the encoder. The initial negative effect may be in part due to the phonological reduction of high-frequency items: the raw acoustics of the high-frequency V+*up* phrases may be more reduced and only after some abstraction does the model begin to represent the *up* in V+*up* phrases more similarly to the standalone word. Interestingly, the trend for predictability is the opposite: it starts out positive and ends up negative, suggesting that the encoder is able to encode high-predictability phrases as containing *up* in early layers, while the encoding of V+*up* phrases becomes agnostic to predictability in later layers.

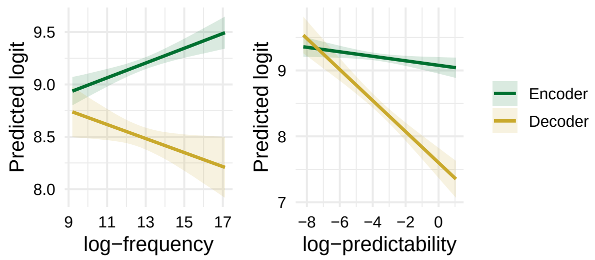

Figure 5: Final-layer brms predicted logit by frequency (left) and predictability (right) for Whisper encoder and decoder. Lines and bands as in Figure 1.

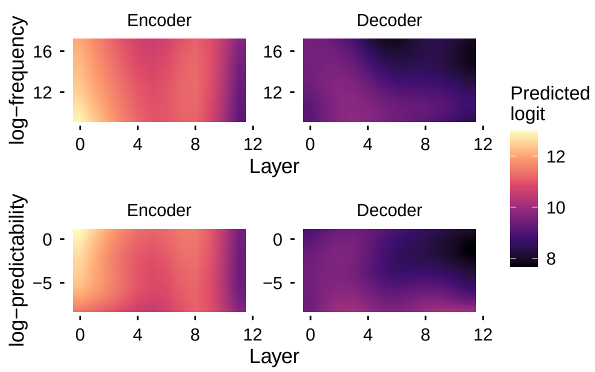

Figure 6: GAM-predicted logit by layer and predictor for Whisper encoder and decoder. Top: frequency; bottom: predictability. Color = logit; darker = lower logit (less similar to standalone *up*); random effects excluded.

5.3 Discussion

The results of Whisper demonstrate an interesting difference between audio speech recognizers and text-based language models: the encoder of ASR models represents high-frequency V+*up* phrases *more* similarly to the standalone *up* representations relative to lower-frequency V+*up* phrases. High-predictability phrases, by contrast, are represented *less* similarly to standalone *up* even at the encoder's final layer, and by the final layer of the decoder, both effects are strongly negative, indicating fully holistic representations at the output stage.

We propose that this pattern reflects the functional division between Whisper's encoder and decoder. The encoder's primary role is acoustic processing, converting the raw audio signal into representations suitable for recognition. Crucially, frequency and predictability are not equivalent dimensions: a phrase can be highly predictable yet low in frequency (*clammed up*, for instance, has a high predictability value, yet occurs over 2,600 times less frequently than *set up*). High-frequency V+*up* phrases appear many times in the acoustic signal; even though frequent phrases tend to undergo phonetic reduction, repeated acoustic exposure gives the encoder extensive experience with *up* in context, anchoring its acoustic representation close to the standalone form. High-predictability phrases are not necessarily high-frequency, so the encoder receives fewer total acoustic exposures to them, and their reduced forms of *up* may be less reliably associated with the standalone *up* representation. Holistic storage effects, by contrast, are most apparent in the decoder, where linguistic rather than acoustic representations are primary.

6 Discussion and Conclusion

The present study demonstrates that in general, both text-based LLMs and ASR models represent high-frequency and high-predictability V+*up* phrases holistically. Further, the effect of predictability is clearer for larger models, suggesting that increasing the number of parameters facilitates predictability effects. Finally, the results suggest that for larger models, these effects emerge in earlier layers, while for smaller models the effect emerges in later layers. The Whisper results add a further nuance: the encoder's positive frequency effect, absent in text-based LLMs, likely reflects the acoustic nature of its representations: repeated encounters with the reduced but recognizable *up* in high-frequency phrases anchor its acoustic representation near the standalone form. Holistic storage effects are instead most pronounced in the decoder, where linguistic rather than acoustic representations are primary, and both frequency and predictability drive representations strongly away from standalone *up*. This suggests that holistic storage is localized to the linguistic decoder in audio models, and that modality shapes not just whether holistic storage emerges, but at which processing stage it does so.

The by-layer results suggest that holistic storage is better understood as a spectrum than a binary distinction. Representations that diverge only in later layers may reflect a form of partial holistic storage, where lower-level representations remain more compositional but higher-level abstractions increasingly integrate item-specific information. Larger models, which have encountered V+*up* phrases more frequently, show more divergent representations even in early hidden layers, suggesting that with sufficient exposure or capacity, storage-like behavior becomes a more pervasive organizational principle.

Holistic storage is itself a form of exemplar-based representation: rather than computing a phrase from its parts each time, the whole phrase is represented as a stored unit. If holistic storage and grammatical abstraction arise from the same exemplar-based process, the computation–storage tradeoff may be a continuum rather than a binary. Transformer models offer a potential computational instantiation of this view: they encode no explicit rules, yet develop frequency- and predictability-graded holistic representations purely from distributional exposure, suggesting that exemplar-based generalization is sufficient to produce the patterns observed in human behavior. If so, the same underlying mechanism may give rise to both holistic storage and grammatical abstraction in human language users.

Overall, the present study demonstrates another dimension upon which LLMs and even ASR models' behavior resembles that of humans, providing further evidence for usage-based theories of holistic storage. More broadly, the results suggest that Transformer architectures may serve as a productive modeling framework for investigating not just whether human-like storage patterns emerge, but *how* and *when* they do so, and at a level of mechanistic detail that behavioral data alone cannot provide.

7 Limitations

The primary limitation is that we examined only one construction (V+*up* phrases) and in one language (English). We look forward to expanding the present study to other languages and other constructions in future work. We also only examined representations at the final checkpoint; however, we look forward to examining the emergence of storage as a function of training dynamics in the future.

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A BabyLM Model Training

Models were trained from random initialization for 20 epochs using the AdamW optimizer with fused weight updates and bfloat16 mixed precision. The learning rate was set to 3×10^{-4} for the 125M model and 1×10^{-4} for the 350M and 1.3B models, each preceded by a linear warmup over 10% of total training steps. Training was distributed across two NVIDIA A100 80GB GPUs. Hyperparameter details are given in Table 1.

Parameter	OPT-125M	OPT-350M	OPT-1.3B
Hidden size	768	1,024	2,048
Attention heads	12	16	32
Layers	12	24	24
FFN intermediate size	3,072	4,096	8,192
Vocabulary size	8,192	8,192	8,192
Context length (tokens)	1,024	1,024	1,024
Batch size (per device)	400	200	100
Learning rate	3×10^{-4}	1×10^{-4}	1×10^{-4}
Warmup steps	366	732	1,465
Training epochs	20	20	20
Optimizer	AdamW (fused)	AdamW (fused)	AdamW (fused)
Precision	bfloat16	bfloat16	bfloat16

Table 1: Hyperparameters for the three BabyLM models. All models share a BPE tokenizer (vocabulary size 8,192) trained on the BabyLM corpus. Training used two NVIDIA A100 80GB GPUs with distributed data parallelism.

B BabyLM Model Convergence

All three models trained to completion over 20 epochs, with training loss decreasing monotonically throughout (Figure 7). Final gradient norms were small for all models (0.16, 0.24, and 0.38 for the 125M, 350M, and 1.3B models, respectively), and the learning rate decayed to zero at the end of training, consistent with full convergence under the scheduled warmup-then-decay regime. No signs of loss divergence or instability were observed.

The 1.3B model achieves the lowest training loss (2.55) and validation perplexity (15.18), as expected given its greater capacity (Table 2). The 350M model, however, shows slightly elevated training loss (2.63) and validation perplexity (18.86) relative to the 125M model (training loss 2.55; validation perplexity 16.81). This is likely attributable to the lower learning rate applied to the 350M model (1×10^{-4} vs. 3×10^{-4} for the 125M), which, combined with a relatively modest dataset size of 150M tokens, may have produced slower effective convergence. The effect is small and does not affect our qualitative comparisons, but we note it here for completeness.

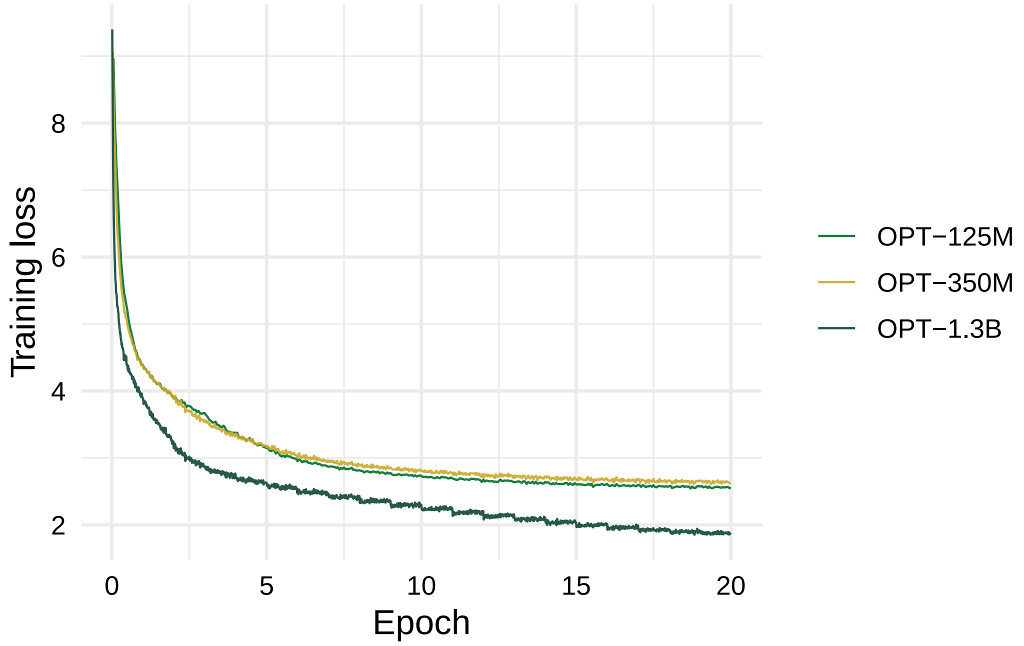

Figure 7: Training loss over 20 epochs for each BabyLM model. Loss is logged every 10 gradient steps.

Model	Val. loss	Perplexity
OPT-125M	2.82	16.81
OPT-350M	2.94	18.86
OPT-1.3B	2.72	15.18

Table 2: Final validation loss and perplexity for each BabyLM model, evaluated on the BabyLM development set after 20 epochs of training.

C Test Set Items

Model	N items	Med. freq	Freq range	Med. predic.	Predic. range
OLMo-3 7B	4081	43608	2–103,842,740	-5.38	-10.98–2.74
BabyLM	1039	920	20–390,086	-3.40	-10.38–1.99
Whisper encoder	1426	243412	960–103,842,740	-4.08	-10.98–2.74
Whisper decoder	1426	243412	960–103,842,740	-4.08	-10.98–2.74

Table 3: Test set statistics for all experiments. Only items with a valid (non-zero) predictability estimate are included. Frequency is the raw corpus count; predictability is the log odds (see Equation 2). BabyLM models (125M, 350M, 1.3B) share identical items and corpus statistics. Whisper encoder and decoder evaluate the same audio items with Dolma-derived frequency counts.

D Experiment 1: UP Independently

Model	Train pos	Train neg	Val pos	Val neg
OLMo-3 7B	1,000	1,000	992	1,000
BabyLM 125M	1,000	1,000	999	1,000
BabyLM 350M	1,000	1,000	999	1,000
BabyLM 1.3B	1,000	1,000	916	1,000

Table 4: Classifier training and validation split sizes for Experiment 1 (UP independently). Positive (pos) = standalone *up*; negative (neg) = other alphabetic token from the same sentence.

Model	Coef	Est.	Err.	95% CI	% > 0
OLMo-3 7B	Intercept	10.29	0.04	[10.22, 10.37]	100.0
OLMo-3 7B	Freq	-0.57	0.04	[-0.65, -0.49]	0.0
OLMo-3 7B	Predic.	-1.64	0.04	[-1.72, -1.56]	0.0
OLMo-3 7B	Freq:Predic.	0.03	0.04	[-0.05, 0.10]	74.5
BabyLM 125M	Intercept	8.72	0.04	[8.65, 8.80]	100.0
BabyLM 125M	Freq	-0.30	0.04	[-0.37, -0.23]	0.0
BabyLM 125M	Predic.	-0.07	0.04	[-0.14, 0.01]	3.2
BabyLM 125M	Freq:Predic.	-0.12	0.04	[-0.20, -0.05]	0.1
BabyLM 350M	Intercept	8.43	0.04	[8.35, 8.50]	100.0
BabyLM 350M	Freq	-0.30	0.04	[-0.38, -0.23]	0.0
BabyLM 350M	Predic.	-0.04	0.04	[-0.11, 0.04]	18.2
BabyLM 350M	Freq:Predic.	-0.16	0.04	[-0.24, -0.08]	0.0
BabyLM 1.3B	Intercept	8.57	0.05	[8.47, 8.68]	100.0
BabyLM 1.3B	Freq	-0.27	0.05	[-0.38, -0.17]	0.0
BabyLM 1.3B	Predic.	-0.20	0.05	[-0.30, -0.10]	0.0
BabyLM 1.3B	Freq:Predic.	-0.14	0.05	[-0.24, -0.03]	0.7

Table 5: Joint frequency × predictability model at the final layer for UP independently. Estimates are from Bayesian linear mixed-effects models (brms); % > 0 indicates the percentage of posterior samples with a positive estimate.

Model	Predictor	EDF	F	p
OLMo-3 7B	Frequency	23.48	39510.17	< 0.001
BabyLM 125M	Frequency	21.52	3529.76	< 0.001
BabyLM 350M	Frequency	21.62	26424.27	< 0.001
BabyLM 1.3B	Frequency	21.00	28461.84	< 0.001
OLMo-3 7B	Predictability	23.44	41912.50	< 0.001
BabyLM 125M	Predictability	22.52	3421.22	< 0.001
BabyLM 350M	Predictability	22.98	25267.09	< 0.001
BabyLM 1.3B	Predictability	22.47	27285.85	< 0.001

Table 6: GAM tensor-product smooth summary for UP independently. EDF = estimated degrees of freedom; p = p-value for the te(predictor, layer) smooth term.

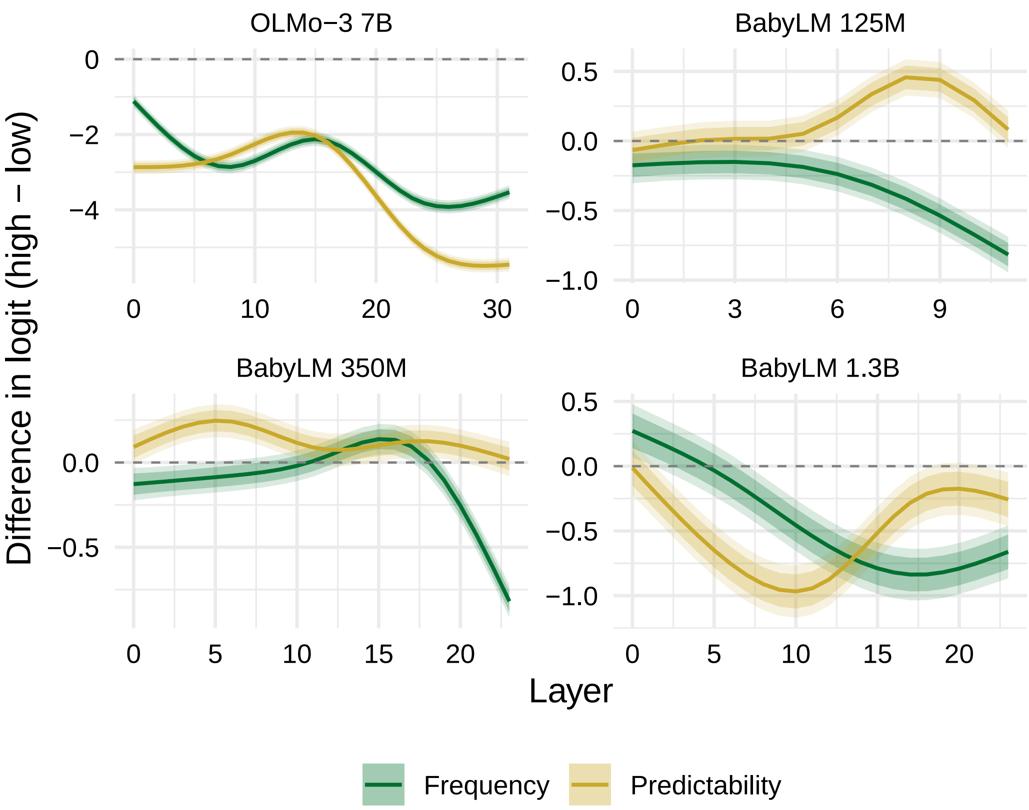

Figure 8: Layer-by-layer difference in GAM-predicted logit between high (90th percentile) and low (10th percentile) predictor values, for UP independently. Green = frequency effect; yellow = predictability effect. Inner and outer shaded ribbons show 80% and 95% confidence intervals derived from the GAM standard error. A negative difference indicates that high-predictor phrases yield a lower classifier logit (i.e., their representation of *up* is less similar to the standalone class).

E Experiment 2: UP as Subword

Model	Train standalone	Train subword	Train neg	Val rows
OLMo-3 7B	1,000	1,000	2,000	3,868
BabyLM 125M	1,000	1,000	2,000	3,582
BabyLM 350M	1,000	1,000	2,000	3,582
BabyLM 1.3B	1,000	1,000	2,000	3,773

Table 7: Classifier training and validation split sizes for Experiment 2 (UP as subword). Positive examples: 1,000 standalone *up* + 1,000 unique up-within-word types. Negative examples: 1,000 from each sentence pool. Val sizes vary by model due to tokenizer coverage.

Model	Coef	Est.	Err.	95% CI	% > 0
OLMo-3 7B	Intercept	12.34	0.04	[12.26, 12.42]	100.0
OLMo-3 7B	Freq	-0.48	0.04	[-0.57, -0.40]	0.0
OLMo-3 7B	Predic.	-1.04	0.04	[-1.12, -0.96]	0.0
OLMo-3 7B	Freq:Predic.	-0.05	0.04	[-0.13, 0.03]	9.3
BabyLM 125M	Intercept	12.26	0.07	[12.13, 12.40]	100.0
BabyLM 125M	Freq	-0.45	0.07	[-0.59, -0.32]	0.0
BabyLM 125M	Predic.	0.11	0.07	[-0.03, 0.24]	93.2
BabyLM 125M	Freq:Predic.	-0.17	0.07	[-0.32, -0.03]	0.9
BabyLM 350M	Intercept	12.70	0.06	[12.57, 12.82]	100.0
BabyLM 350M	Freq	-0.48	0.06	[-0.60, -0.36]	0.0
BabyLM 350M	Predic.	0.00	0.06	[-0.12, 0.13]	52.2
BabyLM 350M	Freq:Predic.	-0.12	0.07	[-0.25, 0.01]	3.4
BabyLM 1.3B	Intercept	10.98	0.05	[10.88, 11.07]	100.0
BabyLM 1.3B	Freq	-0.22	0.05	[-0.31, -0.12]	0.0
BabyLM 1.3B	Predic.	-0.07	0.05	[-0.17, 0.02]	6.2
BabyLM 1.3B	Freq:Predic.	-0.04	0.05	[-0.14, 0.07]	24.5

Table 8: Joint frequency × predictability model at the final layer for UP as subword. Estimates are from Bayesian linear mixed-effects models (brms); % > 0 indicates the percentage of posterior samples with a positive estimate.

Model	Predictor	EDF	F	p
OLMo-3 7B	Frequency	23.21	52988.02	< 0.001
BabyLM 125M	Frequency	18.33	543.59	< 0.001
BabyLM 350M	Frequency	21.44	6564.50	< 0.001
BabyLM 1.3B	Frequency	20.68	20878.03	< 0.001
OLMo-3 7B	Predictability	23.13	54126.21	< 0.001
BabyLM 125M	Predictability	22.15	517.34	< 0.001
BabyLM 350M	Predictability	22.30	6379.07	< 0.001
BabyLM 1.3B	Predictability	21.51	20339.52	< 0.001

Table 9: GAM tensor-product smooth summary for UP as subword. EDF = estimated degrees of freedom; p = p-value for the te(predictor, layer) smooth term.

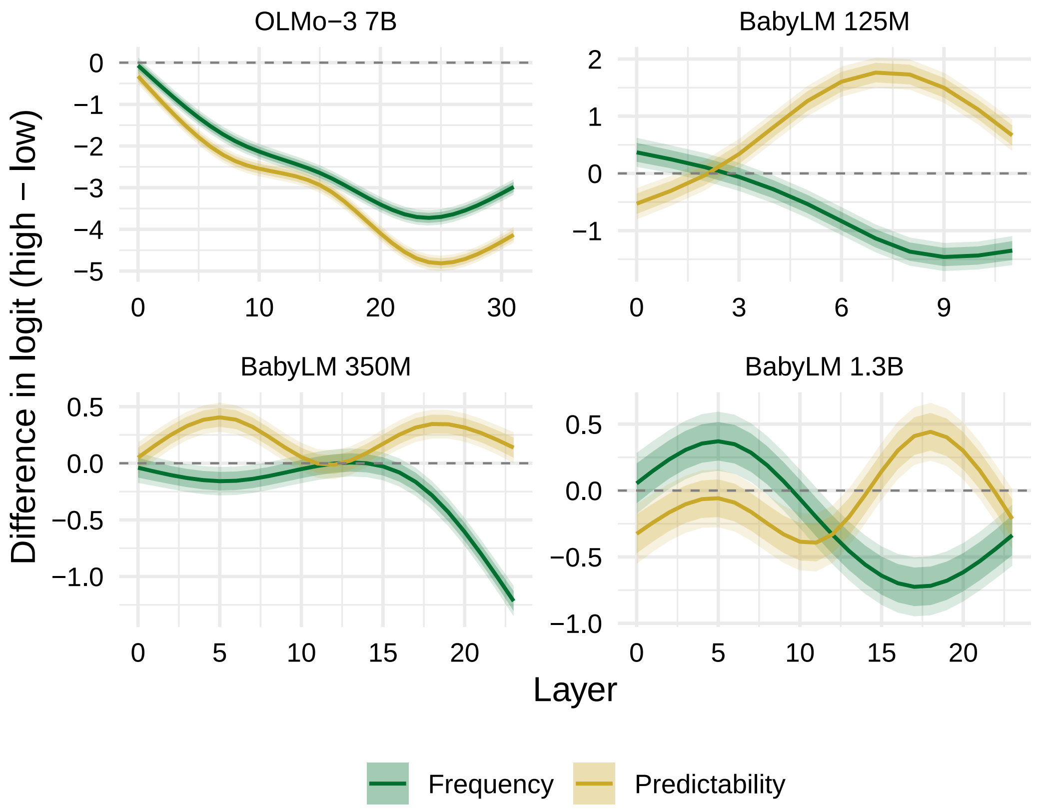

Figure 9: Layer-by-layer difference in GAM-predicted logit between high (90th percentile) and low (10th percentile) predictor values, for UP as subword. Green = frequency effect; yellow = predictability effect. Inner and outer shaded ribbons show 80% and 95% confidence intervals derived from the GAM standard error. A negative difference indicates that high-predictor phrases yield a lower classifier logit (i.e., their representation of *up* is less similar to the standalone class).

F Experiment 3: Whisper

Component	Train pos	Train neg	Val pos	Val neg
Encoder	1,000	1,000	1,000	1,000
Decoder	1,000	1,000	1,000	1,000

Table 10: Classifier training and validation split sizes for Experiment 3 (Whisper). Positive examples are *word_up* instances; negatives are random non-*up* alphabetic tokens from the same segment.

Model	Coef	Est.	Err.	95% CI	% > 0
Encoder	Intercept	9.21	0.03	[9.15, 9.26]	100.0
Encoder	Freq	0.12	0.03	[0.06, 0.17]	100.0
Encoder	Predic.	-0.07	0.03	[-0.12, -0.01]	1.3
Encoder	Freq:Predic.	0.01	0.03	[-0.05, 0.06]	63.1
Decoder	Intercept	8.50	0.05	[8.39, 8.60]	100.0
Decoder	Freq	-0.11	0.05	[-0.21, -0.00]	2.2
Decoder	Predic.	-0.45	0.05	[-0.56, -0.34]	0.0
Decoder	Freq:Predic.	-0.13	0.05	[-0.23, -0.03]	0.5

Table 11: Joint frequency × predictability model at the final layer for Whisper encoder and decoder. Estimates are from Bayesian linear mixed-effects models (brms); % > 0 indicates the percentage of posterior samples with a positive estimate.

Model	Predictor	EDF	F	p
Encoder	Frequency	13.12	521.72	< 0.001
Decoder	Frequency	22.14	543.35	< 0.001
Encoder	Predictability	18.98	377.97	< 0.001
Decoder	Predictability	22.35	562.36	< 0.001

Table 12: GAM tensor-product smooth summary for Whisper encoder and decoder. EDF = estimated degrees of freedom; p = p-value for the te(predictor, layer) smooth term.

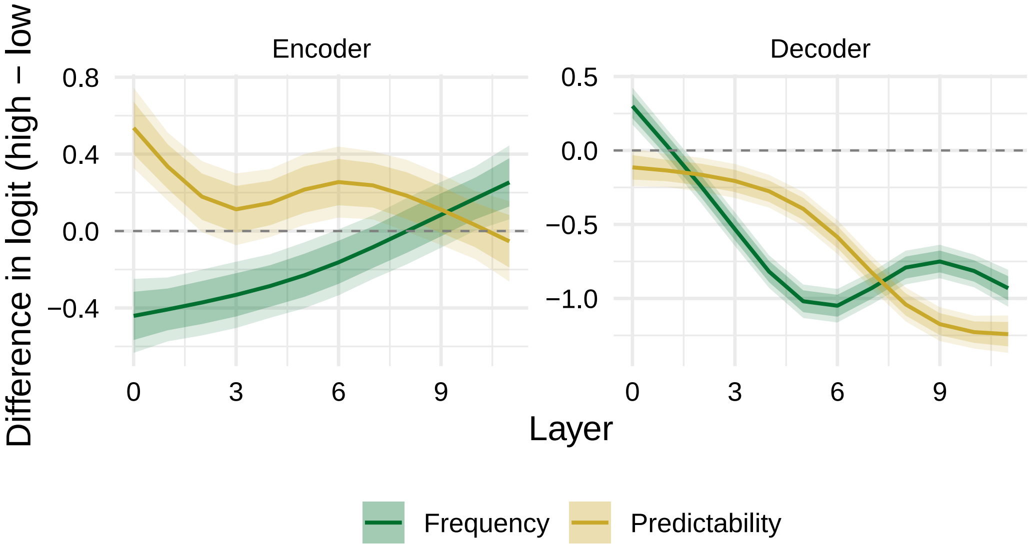

Figure 10: Layer-by-layer difference in GAM-predicted logit between high (90th percentile) and low (10th percentile) predictor values, for Whisper encoder and decoder. Green = frequency effect; yellow = predictability effect. Inner and outer shaded ribbons show 80% and 95% confidence intervals derived from the GAM standard error. A negative difference indicates that high-predictor phrases yield a lower classifier logit (i.e., their representation of *up* is less similar to the standalone class).